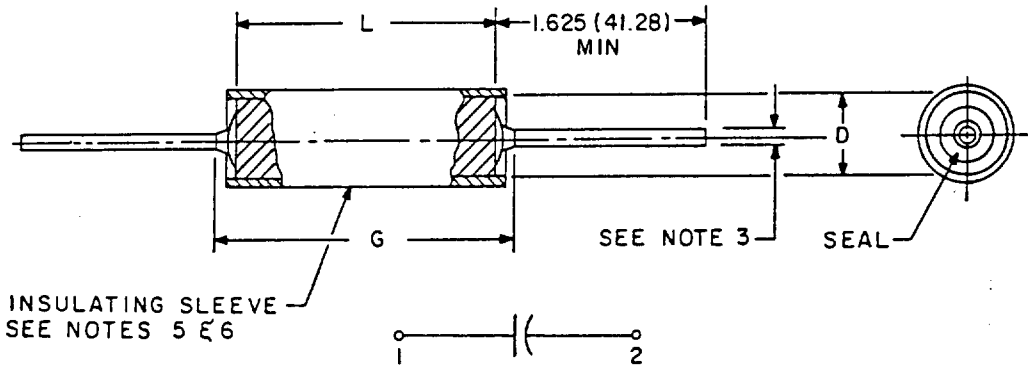
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, SUPERMETALLIZED PLASTIC FILM DIELECTRIC, (DC, AC, OR DC AND AC), HERMETICALLY SEALED IN METAL CASES, ESTABLISHED RELIABILITY, STYLES CRH11, CRH12 AND CRH13 (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-83421.



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses and are given for general information only.
3. Number 18 AWG wire .040 inch ±.002 (1.02 ±0.05 mm).
4. See table I for additional dimensions.
5. Insulating sleeve shall extend beyond the capacitor body. Insulating sleeve thickness shall not exceed .005 inch (0.13 mm).
6. Plastic insulating sleeve shall be transparent; marking shall be applied to the capacitor case.

FIGURE 1. Capacitor configuration.

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TABLE I. Electrical characteristics, dimensions, and dash numbers.

Capacitance value (nom) (in µF)	Dimensions 1/ (in inches with mm in parentheses) L +.030(0.76) -.090(2.29)	D +.020(0.51) -.010(0.25)	G (max)	Dash number 2/ Capacitance tolerance value (in %) ±0.25	±0.5	±1.0	±2.0	±5.0	±10.0	ESR 20 kHz 100 kHz (ohms maximum)	AC rating maximum 400 Hz	Ripple current 20 kHz to 100 kHz (amperes rms) maximum case temperature °C 3/ 25	35	45	55	65	75	85	95	105
CRH11 - 100 volts (dc rating)																				
0.47	0.875(22.23)	0.400(10.16)	1.075(27.31)	1001-	1002-	1003-	1004-	1005-	1006-	0.025	60 volts rms	5.3	4.9	4.6	4.2	3.7	3.2	2.6	1.9	0.8
0.56	"	0.500(12.70)	"	1007-	1008-	1009-	1010-	1011-	1012-	0.024	"	6.0	5.6	5.2	4.8	4.3	3.7	3.0	2.1	1.0
0.68	"	"	"	1013-	1014-	1015-	1016-	1017-	1018-	0.023	"	6.2	5.8	5.4	4.9	4.4	3.8	3.1	2.2	1.0
0.82	"	"	"	1019-	1020-	1021-	1022-	1023-	1024-	0.022	"	6.3	5.9	5.5	5.0	4.5	3.9	3.2	2.2	1.0
1.0	0.906(23.01)	0.562(14.27)	1.106(28.09)	1025-	1026-	1027-	1028-	1029-	1030-	0.017	"	7.7	7.2	6.7	6.1	5.4	4.7	3.9	2.7	1.2
2.0	1.094(27.80)	0.670(17.02)	1.294(32.87)	1031-	1032-	1033-	1034-	1035-	1036-	0.014	"	10.3	9.7	8.9	8.2	7.3	6.3	5.2	3.7	1.6
3.0	1.094(27.80)	0.750(19.05)	1.294(32.87)	1037-	1038-	1039-	1040-	1041-	1042-	0.013	"	11.4	10.7	9.9	9.0	8.1	7.0	5.7	4.0	1.8
5.0	1.406(35.70)	0.750(19.05)	1.606(40.79)	1043-	1044-	1045-	1046-	1047-	1048-	0.012	"	13.6	12.7	11.8	10.7	9.6	8.3	6.8	4.8	2.1
10.0	1.687(42.85)	1.000(25.40)	1.887(47.93)	1049-	1050-	1051-	1052-	1053-	1054-	0.010	"	15.0	15.0	15.0	14.3	12.8	11.1	9.1	6.4	2.9
20.0	2.437(61.90)	1.000(25.40)	2.637(66.98)	1055-	1056-	1057-	1058-	1059-	1060-	0.009	"	15.0	15.0	15.0	15.0	15.0	14.1	11.5	8.2	3.6
CRH12 - 200 volts (dc rating)																				
0.18	0.875(22.23)	0.400(10.16)	1.075(27.31)	2001-	2002-	2003-	2004-	2005-	2006-	0.031	120 volts rms	4.8	4.5	4.1	3.8	3.4	2.9	2.4	1.7	0.8
0.22	"	0.500(12.70)	"	2007-	2008-	2009-	2010-	2011-	2012-	0.030	"	5.4	5.1	4.7	4.3	3.8	3.3	2.7	1.9	0.9
0.27	"	"	"	2013-	2014-	2015-	2016-	2017-	2018-	0.029	"	5.5	5.2	4.8	4.4	3.9	3.4	2.8	2.0	0.9
0.33	"	"	"	2019-	2020-	2021-	2022-	2023-	2024-	0.028	"	5.6	5.3	4.9	4.5	4.0	3.5	2.8	2.0	0.9
0.39	0.906(23.01)	0.562(14.27)	1.106(28.09)	2025-	2026-	2027-	2028-	2029-	2030-	0.026	"	6.2	5.8	5.4	4.9	4.4	3.8	3.1	2.2	1.0
0.47	1.094(27.80)	"	1.294(32.87)	2031-	2032-	2033-	2034-	2035-	2036-	0.025	"	7.0	6.5	6.1	5.5	4.9	4.3	3.5	2.5	1.1
0.56	"	"	"	2037-	2038-	2039-	2040-	2041-	2042-	0.024	"	7.2	6.7	6.2	5.7	5.1	4.4	3.6	2.5	1.1
0.68	"	0.670(17.02)	"	2043-	2044-	2045-	2046-	2047-	2048-	0.023	"	8.0	7.5	6.9	6.3	5.7	4.9	4.0	2.8	1.3
0.82	"	"	"	2049-	2050-	2051-	2052-	2053-	2054-	0.022	"	8.2	7.7	7.1	6.5	5.8	5.0	4.1	2.9	1.3
1.0	"	"	"	2055-	2056-	2057-	2058-	2059-	2060-	0.021	"	8.4	7.9	7.3	6.7	6.0	5.2	4.2	3.0	1.3
2.0	1.406(35.70)	0.750(19.05)	1.606(40.79)	2061-	2062-	2063-	2064-	2065-	2066-	0.017	"	11.1	10.4	9.6	8.8	7.8	6.8	5.5	3.9	1.8
3.0	1.437(36.50)	1.000(25.40)	1.637(41.58)	2067-	2068-	2069-	2070-	2071-	2072-	0.015	"	13.9	13.0	12.0	11.0	9.8	8.5	7.0	4.9	2.2
5.0	1.687(42.85)	1.000(25.40)	1.887(47.93)	2073-	2074-	2075-	2076-	2077-	2078-	0.013	"	15.0	15.0	14.2	12.9	11.6	10.0	8.2	5.8	2.6
CRH13 - 400 volts (dc rating)																				
0.056	0.875(22.23)	0.400(10.16)	1.075(27.31)	3001-	3002-	3003-	3004-	3005-	3006-	0.058	240 volts rms	3.5	3.3	3.0	2.8	2.5	2.1	1.7	1.2	0.6
0.068	"	0.500(12.70)	"	3007-	3008-	3009-	3010-	3011-	3012-	0.046	"	4.4	4.1	3.8	3.5	3.1	2.7	2.2	1.5	0.7
0.082	"	"	"	3013-	3014-	3015-	3016-	3017-	3018-	0.039	"	4.7	4.4	4.1	3.7	3.4	2.9	2.4	1.7	0.7
0.10	"	"	"	3019-	3020-	3021-	3022-	3023-	3024-	0.035	"	5.0	4.7	4.4	4.0	3.6	3.1	2.5	1.8	0.8
0.12	0.906(23.01)	0.562(14.27)	1.106(28.09)	3025-	3026-	3027-	3028-	3029-	3030-	0.033	"	5.5	5.2	4.8	4.4	3.9	3.4	2.8	2.0	0.9
0.15	"	0.670(17.02)	"	3031-	3032-	3033-	3034-	3035-	3036-	0.032	"	6.2	5.8	5.3	4.9	4.4	3.8	3.1	2.2	1.0
0.18	"	"	"	3037-	3038-	3039-	3040-	3041-	3042-	0.031	"	6.3	5.9	5.4	5.0	4.4	3.8	3.1	2.2	1.0
0.22	1.094(27.80)	"	1.294(32.87)	3043-	3044-	3045-	3046-	3047-	3048-	0.030	"	7.0	6.6	6.1	5.6	5.0	4.3	3.5	2.5	1.1
0.27	"	"	"	3049-	3050-	3051-	3052-	3053-	3054-	0.029	"	7.2	6.7	6.2	5.7	5.1	4.4	3.6	2.5	1.1
0.33	"	"	"	3055-	3056-	3057-	3058-	3059-	3060-	0.028	"	7.3	6.8	6.3	5.8	5.2	4.5	3.7	2.6	1.2
0.39	"	0.750(19.05)	"	3061-	3062-	3063-	3064-	3065-	3066-	0.026	"	7.9	7.4	6.8	6.2	5.6	4.8	3.9	2.8	1.2
0.47	"	"	"	3067-	3068-	3069-	3070-	3071-	3072-	0.025	"	8.1	7.6	7.0	6.4	5.7	4.9	4.0	2.9	1.3
0.56	1.406(35.70)	"	1.606(40.79)	3073-	3074-	3075-	3076-	3077-	3078-	0.025	"	9.2	8.6	7.9	7.2	6.5	5.6	4.6	3.2	1.4
0.68	1.406(35.70)	"	1.606(40.79)	3079-	3080-	3081-	3082-	3083-	3084-	0.024	"	9.4	8.8	8.1	7.4	6.6	5.7	4.7	3.3	1.5
0.82	1.656(42.06)	"	1.856(47.14)	3085-	3086-	3087-	3088-	3089-	3090-	0.023	"	10.4	9.7	9.0	8.2	7.4	6.4	5.2	3.7	1.6
1.0	1.656(42.06)	"	1.856(47.14)	3091-	3092-	3093-	3094-	3095-	3096-	0.022	"	10.7	10.0	9.3	8.5	7.6	6.5	5.3	3.8	1.7
2.0	1.938(49.23)	1.000(25.40)	2.138(54.31)	3097-	3098-	3099-	3100-	3101-	3102-	0.017	"	15.0	14.1	13.0	11.9	10.6	9.2	7.5	5.3	2.4

1/ L and D are bare case dimensions (see figure 1).

2/ The complete dash number will include the applicable letter completing the FR level symbol (M = M, P = P, etc.).

3/ This is the ambient case temperature prior to the application of current.

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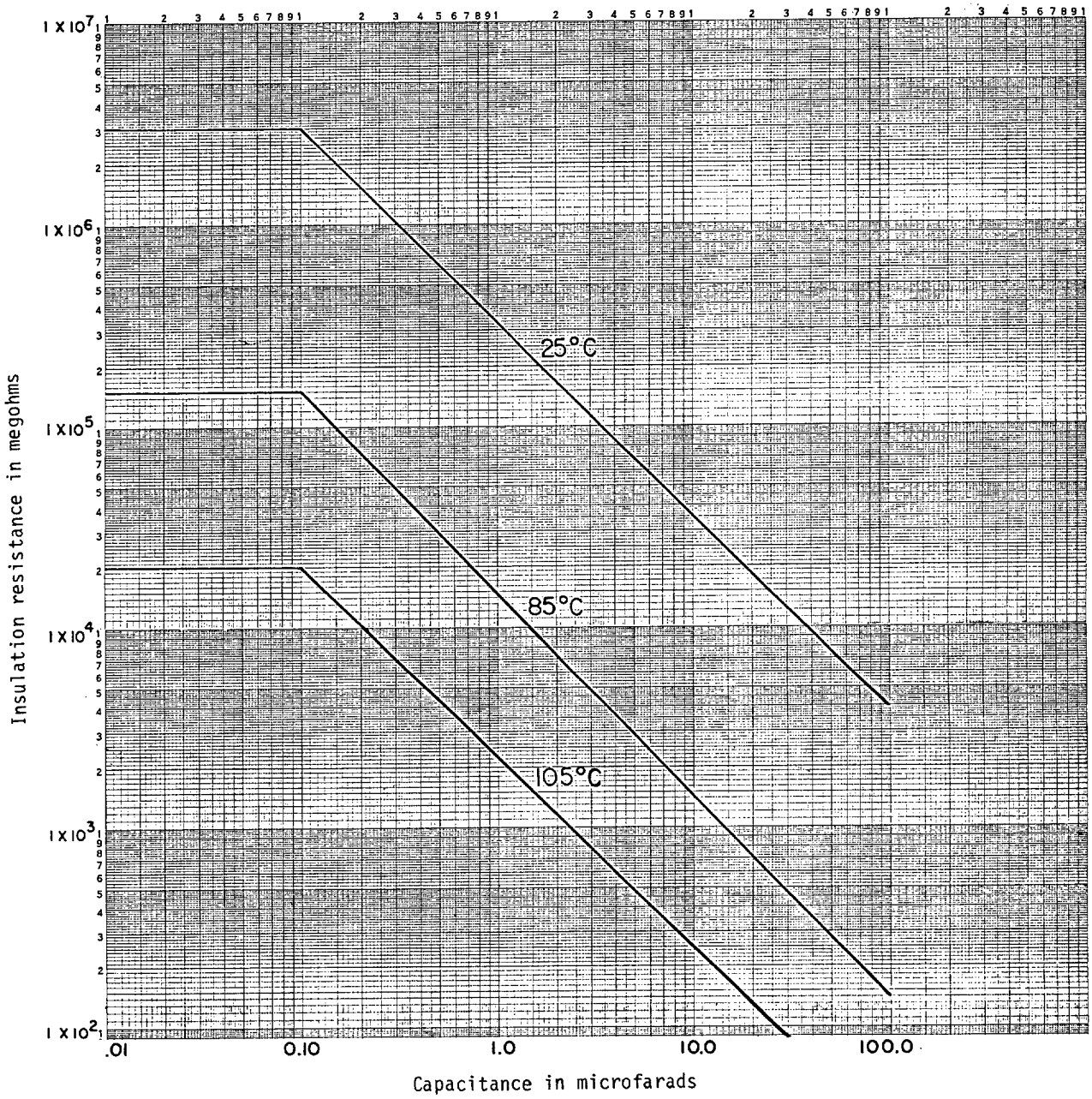


FIGURE 2. Insulation resistance vs capacitance value.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Dielectric: Polypropylene, metallized.

Case material: Nonmagnetic.

Leads: Type designation in accordance with MIL-STD-1276.

L W (1.02 ±0.05) W[1/] 4 N.

Capacitance (cap) value: See table I.

Capacitance (cap) tolerance: See table I.

DC rated voltage: 100, 200, and 400 V dc.

Operating temperature: -55°C to +105°C.

AC current ratings: See table I.

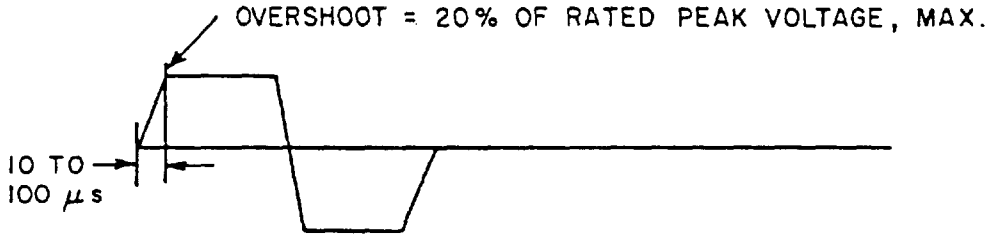
Failure rate (FR) levels: M (1.0%), P (0.1%), R (0.01%), and S (0.001%).

Thermal shock: 10 cycles, method 107 of MIL-STD-202, -65°C, +105°C.

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIa, followed by test condition B. Leakage rate sensitivity 10^{-6} atm cm^3/s.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202, terminal to terminal.

AC: 100 Hz ±10 Hz square wave, peak-to-peak voltage, 3 times dc rated voltage for 60 to 90 seconds, not to exceed 800 V p/p.



DC: 200 percent of dc rated voltage for 60 seconds minimum.

Insulation resistance (IR): Method 302 of MIL-STD-202. Charge to rated voltage, +105°C for 5 minutes maximum; however, for capacitance values greater than 1.0 μF, an additional 1 minute per μF is permitted.

1/ .562 inch and larger diameters: Type C.

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Terminal to terminal: See figure 2.

Dissipation factor (DF): 0.15 percent maximum at 1 kHz.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet). 125 percent of dc rated voltage applied. See MIL-C-83421 for voltage limitations.

Burn-in: 140 percent of dc rated voltage shall be applied for 48 hours minimum at +105°C +4°C, -0°C.

Thermal aging: In accordance with MIL-C-83421, 10 cycles.

Thermal shock: Method 107 of MIL-STD-202, condition B, 10 cycles.

Dielectric absorption: 0.1 percent maximum.

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Insulating sleeves: Greater than 4,000 volts dc.

Terminal to terminal: 200 percent of dc rated voltage.

Terminal to case: 200 percent of dc rated voltage.

IR:

Insulating sleeves: 10,000 megohms minimum.

Terminal to terminal: Not less than 50 percent of initial requirement.

Terminal to case: 5,000 megohms minimum.

ΔCap: Maximum of ±0.25 percent.

DF: 0.15 percent.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, ΔCap, and DF: Same as for immersion.

Low temperature life: In accordance with MIL-C-83421, except measurements shall be made at the following temperatures:

t_1 = +25°C: Record initial capacitance dissipation factor and insulation resistance.

t_2 = -65°C: 48 hours rated dc voltage applied.

t_3 = Maximum operating temperature: Two hours rated dc voltage applied (allow for derating).

t_4 = +25°C: Record capacitance, dissipation factor and insulation resistance. Limits of change shall be as specified in table III.

TABLE III. Post measurement limits.

t_1 to t_4	Styles CRH11 through CRH13
Capacitance	±0.5%
Dissipation factor	0.15% maximum
Insulation resistance	Within initial requirement

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Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

ΔCap: Maximum of ±0.25 percent.

Temperature coefficient: In accordance with MIL-C-83421. Capacitance shall be measured at the temperatures specified in table IV.

TABLE IV. Temperature coefficient.

Temperature 1/	Styles CRH11 through CRH13
t_3 to t_2	0 to -4%
t_3 to t_4	0 to +3%
t_3 to t_1 or t_5	±.25%

1/ t_1 = +25°C ±3°C
t_2 = +105°C ±3°C
t_3 = +25°C ±3°C
t_4 = -55°C +0°C, -3°C
t_5 = +25°C ±3°C

DC life (at +105°C): Method 108 of MIL-STD-202.

Accelerated condition: 140 percent of dc rated voltage.

Rated condition: 100 percent of dc rated volage.

IR:

Terminal to terminal: Not less than 33.3 percent of initial requirement.

Terminal to case: 5,000 megohms minimum.

ΔCap: Maximum of ±2.0 percent.

DF (at +25°C) after life: Maximum of 0.25 percent.

AC conditioning: In accordance with MIL-C-83421 with the following exceptions:

The AC performance shall be at the rated ac ripple current at +85°C, 40 kHz for 250 hours, +40, -0 hours (see table V).

TABLE V. AC test samples.

DC rated voltage (volts)	Sample size		Cap range for selection of sample (μF)
	Qualification	Quality conformance, group B	
100	.47μF - 4 20.0 μF - 4	15	.47 to 20.0
200	.18μF - 4 5.0μF - 4	15	.18 to 5.0
400	.056 μF - 4 2.0μF - 4	15	.056 to 2.0

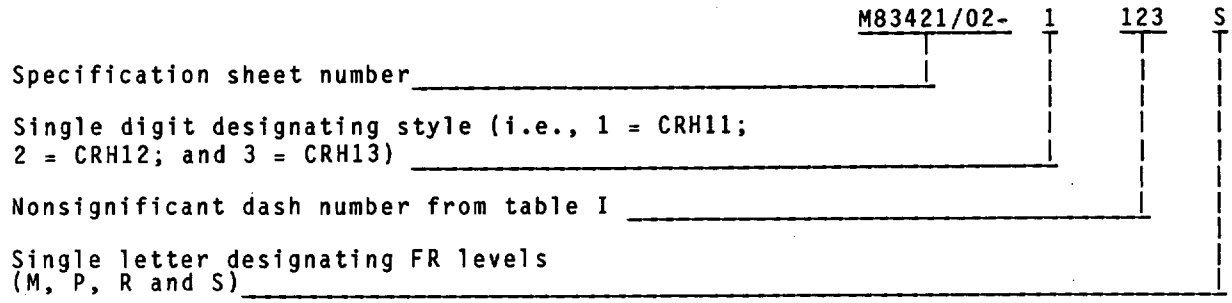
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Qualification: Four each of the highest and lowest capacitance values in each voltage group (see table V).

Quality conformance, group B: 15 units shall be selected which are representative of capacitor ratings produced during the current production period on a semi-annual basis and shall be representative of ratings manufactured during the period (see table V).

Marking: In accordance with MIL-C-83421.

Part number: Consists of the basic number of this specification sheet with a dash number coded as shown in the following:



Packaging: Capacitors will be furnished in tape and reel packaging when so specified in the ordering data.

Custodians:
Army - ER
Navy - EC
Air Force - 11
NASA - NA

Review activities:
Army - AR
Air Force - 17, 85
DLA - ES

User activities:
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Air Force - 11

Agent:
DLA - ES

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